MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SAU
(INCLUDING VARIATION LS1SA)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the cable described herein shall consist of this document and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS1SAU (unarmored) and LS1SA (armored).

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, in accordance with ASTM B 286, size 20-7, uncoated. Maximum diameter 0.039 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Minimum thickness 0.015 inch.
Fourth - Braided shield of AWG no. 36 uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.
- Diameter over completed conductors 0.110 inch nominal.
Sixth - The required number of singles (see table I) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1 from center outward.
Seventh - Binder tape applied helically with overlap.

FSC 6145

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:11Z
Check the source to verify that this is the current version before use.

Construction variant for type LS1SAU

Eighth - Cross-linked polyolefin jacket. Minimum thickness 0.050 inch. Cable surface marking required.

Construction variant for type LS1SA

Eighth - Cross-linked polyolefin jacket. Minimum thickness 0.050 inch.
Ninth - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/41	Size 2/	Number of conductors	Overall diameter (inches)			
			Type LS1SAU		Type LS1SA	
			minimum	maximum	minimum	maximum
-01UN	44	44	0.910	0.990	0.960	1.040

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS1SAU and LS1SA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	10.71
Voltage withstand - volts, root mean square, minimum	
Conductor to shield.....	2000
Shield to shield.....	500
Insulation resistance - megohms/1000 feet, minimum.....	500
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Capacitance - pF/ft at 1 kHz, maximum.....	72
Group B:	
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160

Source: https://assist.dla.mil -- Downloaded: 202640-20T00:11Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - Percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) type LSISAU only - cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Armor - conformance to material, construction and coverage, type LSISA only	No failure
Shield - conformance to material, construction and coverage	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket/insulation)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	--
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:11Z
Check the source to verify that this is the current version before use.

UNIT ORDERING LENGTH: 1000 feet (nominal)

Custodians:
Army - MI
Navy - SH
Air Force - 85

Review activities:
Army - AV, CR, ER
Navy - EC
Air Force - 99
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0958)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:11Z
Check the source to verify that this is the current version before use.